See it working first: the 3D simulation

saral-sim.pages.dev ↗

Scan or click. Any browser, no sign-in. Start with “Start the tour” (12 short chapters). **It is a simulation of our prototype:** example outputs are prepared by the team and metrics are labelled Simulated until measured on the Pi.

WHY SARAL STANDS APART

Not another offline summariser: a lesson built for one learner group

1 A specific learner, not “students”

Designed around learners with Down syndrome and Easy-to-Read principles: short sentences, one idea each, concrete words.

2 A lesson, not just text

Key word and a verified pictogram per sentence, on a large-print screen with one idea per screen. The screen is core, not a bonus.

3 Trust by construction

AI proposes, code verifies. Plain-code checks and a safe fallback mean three verified questions always appear. We state what we do not claim.

4 You can judge it before the event

A full 3D simulation of the device, offline proof and checks, with honest Simulated/Measured labels, and private by design: nothing leaves the device.

THE BARRIER

Standard study text uses long, abstract sentences with no picture support, and an easy-read version of a given passage is **not available on demand**, least of all offline.

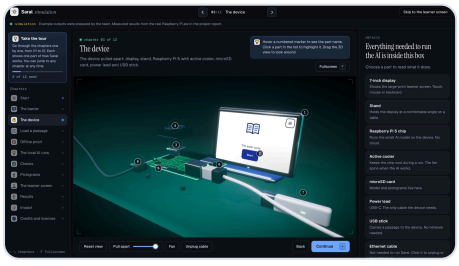
THE LEARNER

People with Down syndrome who benefit from simplified text and visual support. Needs **differ widely**, so Saral adjusts text size and contrast instead of assuming one reading level.

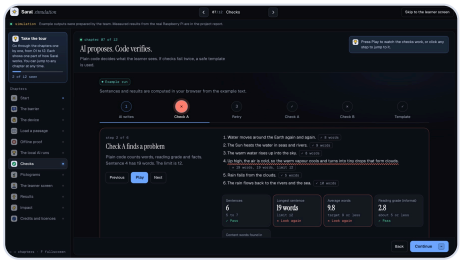
HOW IT WORKS: EVERYTHING RUNS ON THE RASPBERRY PI

Passage 150–300 words → **AI: easy-read** 5–7 sentences → **Check A** length, level, faithful → **AI: questions** 3+ yes/no → **Check B** answers valid → **Pictograms** local library → **Learner screen** large print

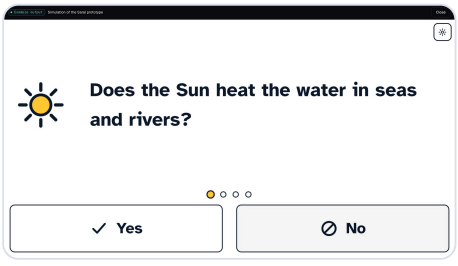
Blue: AI model, on device through ExecuTorch, no cloud inference. Green: deterministic code checks. A failed check is retried up to twice, then a safe fallback is used.



3D device you can orbit, pull apart and unplug



Check A catches a 19-word sentence; retry fixes it



Learner screen: big type, two big answers

THE FIVE MANDATORY DELIVERABLES

REQUIREMENT	HOW SARAL MEETS IT
1 · Barrier, learners, approach	Standard text is hard to read and has no visual support; easy-read text with pictograms for learners with Down syndrome, on the device.
2 · Passage, approved model, ExecuTorch on Pi	Quantized small instruct model via ExecuTorch (XNNPACK), run on the organiser's passage. Final model subject to organiser approval and compatibility testing.
3 · Summary + 3 quiz questions	5–7 easy-read sentences and 3–4 yes/no or two-choice questions. Terminal output always works; the screen is an extra.
4 · No cloud; time, model, size	Wi-Fi off, Ethernet unplugged, offline-check script. Time per stage, model and file size, peak memory with and without the screen.
5 · Impact example	A before/after sentence from the organiser's passage with word counts and the picture used. Example: 16 words and two ideas become 8 words and one idea, with a picture.

How we measure on the Pi

- Model, quantization, file size; peak memory
- Time per stage and total, tokens per second
- Sentence length, reading grade before and after
- Check pass rate, fallback use, pictogram coverage
- Warm-up, then **3 runs** per passage, as ranges; **5+ passages**; two-person quality review

Model and runtime (examples to confirm)

- **Primary:** Llama 3.2 1B Instruct
- **Backup:** Qwen3 0.6B or SmolLM2
- Gemma 3 1B only if its ExecuTorch route is verified
- Quantization scheme finalised after testing
- To confirm: approved list, Pi model, RAM, OS, display

What we do not claim

- **No learning-outcome claims.** Not tested with learners; a 2026 systematic review found Easy Read comprehension evidence limited.
- **Not certified Easy-to-Read.** Needs review by people with intellectual disabilities (planned).
- Checks are rough indicators and cannot catch every meaning error. We report failures.

Plan

- **By 6 Nov:** submit; ask about models, Pi, display
- **To 10 Nov:** pipeline, checks, pictograms, screen
- **11–13 Nov, if shortlisted:** Pi benchmarks
- **14 Nov, NASSCOM:** live run and offline proof
- **Next:** Kannada, read-aloud, educator review

Team

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References

- Bird, G. Helping children with Down's syndrome to learn. *SEN Magazine* 80.
- Inclusion Europe. *Information for all*: European standards for easy-to-read information.
- UK DHSC. Down syndrome needs profile paper, 2025.
- Wilson, H. et al. *JARID* 39(1), 2026 (Easy Read systematic review).

The fastest way to understand Saral is to use it

Open the simulation: saral-sim.pages.dev ↗

Simulation pictograms are original drawings; the device will use an open, licence-checked set.